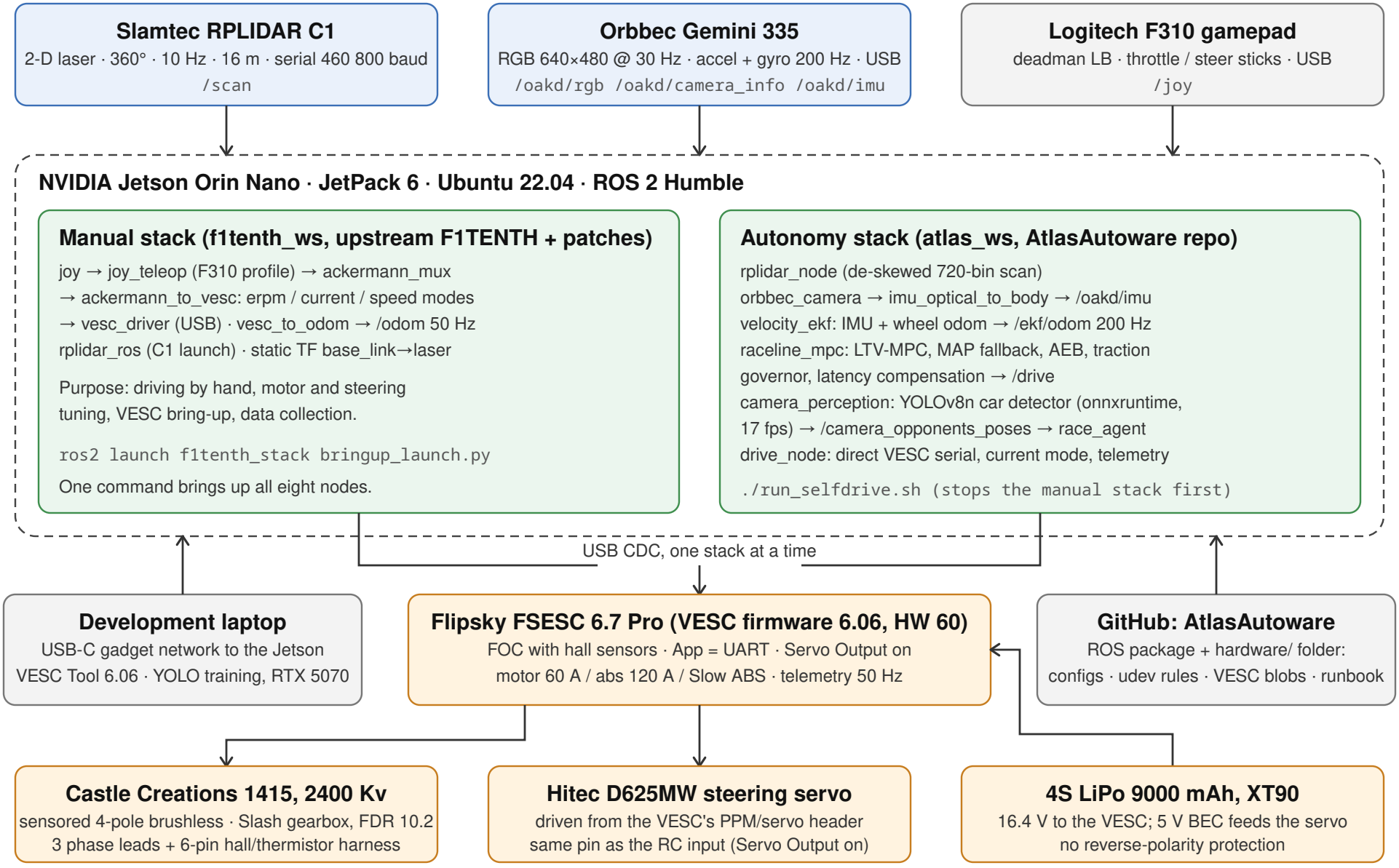

Slamtec RPLIDAR C1
2-D laser · 360° · 10 Hz · 16 m · serial 460 800 baud
/scan
Orbbec Gemini 335
RGB 640×480 @ 30 Hz · accel + gyro 200 Hz · USB
/oakd/rgb /oakd/camera_info /oakd/imu
Logitech F310 gamepad
deadman LB · throttle / steer sticks · USB
/joy
NVIDIA Jetson Orin Nano · JetPack 6 · Ubuntu 22.04 · ROS 2 Humble
Manual stack (f1tenth_ws, upstream F1TENTH + patches)
joy → joy_teleop (F310 profile) → ackermann_mux
→ ackermann_to_vesc: erpm / current / speed modes
→ vesc_driver (USB) · vesc_to_odom → /odom 50 Hz
rplidar_ros (C1 launch) · static TF base_link→laser
Purpose: driving by hand, motor and steering
tuning, VESC bring-up, data collection.
ros2 launch f1tenth_stack bringup_launch.py
One command brings up all eight nodes.
Autonomy stack (atlas_ws, AtlasAutoware repo)
rplidar_node (de-skewed 720-bin scan)
orbbec_camera → imu_optical_to_body → /oakd/imu
velocity_ekf: IMU + wheel odom → /ekf/odom 200 Hz
raceline_mpc: LTV-MPC, MAP fallback, AEB, traction
governor, latency compensation → /drive
camera_perception: YOLOv8n car detector (onnxruntime,
17 fps) → /camera_opponents_poses → race_agent
drive_node: direct VESC serial, current mode, telemetry
./run_selfdrive.sh (stops the manual stack first)
USB CDC, one stack at a time
Development laptop
USB-C gadget network to the Jetson
VESC Tool 6.06 · YOLO training, RTX 5070
Flipsky FSESC 6.7 Pro (VESC firmware 6.06, HW 60)
FOC with hall sensors · App = UART · Servo Output on
motor 60 A / abs 120 A / Slow ABS · telemetry 50 Hz
GitHub: AtlasAutoware
ROS package + hardware/ folder:
configs · udev rules · VESC blobs · runbook
Castle Creations 1415, 2400 Kv
sensored 4-pole brushless · Slash gearbox, FDR 10.2
3 phase leads + 6-pin hall/thermistor harness
Hitec D625MW steering servo
driven from the VESC's PPM/servo header
same pin as the RC input (Servo Output on)
4S LiPo 9000 mAh, XT90
16.4 V to the VESC; 5 V BEC feeds the servo
no reverse-polarity protection